Raul Mendy

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EXPERIENCE

Software Development Engineer II | Volico October 2024 – Present

- Developed an AI-powered customer service agent using a RAG vector database and LLM with speech-to-text and text-to-speech capabilities, enabling automated resolution of common support requests following a 400-client contract expansion — reducing inbound calls to human agents by 60%.
- Designed and implemented an internal infrastructure monitoring platform that ingested real-time data from building management system (BMS) devices via Modbus protocol
- Replaced a deprecated AT&T email-to-SMS service with a self-hosted SMTP routing system that delivered alerts to on-call personnel.

Software Development Engineer I | Florida Power & Light August 2023 – September 2024

- Led the design, development, and deployment of a cloud-based framework supporting smart-grid diagnostics accessible through an integrated chatbot
- Deployed AI models for predictive analysis and natural language processing (NLP), ensuring scalable and efficient data processing.

Research Associate | Florida Atlantic University May 2023 – August 2023

- Developed LSF model and interface for multimodal data analysis, enhancing Prognostic Health Management of smart grid power transformers, achieving 97% fault classification accuracy.

SKILLS

Languages: Python, C++, C, JavaScript, Java, SQL
Backend: FastAPI, Microservices, PostgreSQL, Redis, Qdrant, Docker, Kubernetes, Rails, Node.js
AI/ML: PyTorch, TensorFlow, OpenCV, RAG, Vector Databases, NLP
Cloud & DevOps: AWS, Google Cloud, GitLab CI/CD, Git, Postman
Protocols & Systems: Modbus TCP, SMTP, WebSockets, Real-Time Processing, Linux

PROJECTS

Smart Wildlife Camera System | Skukuza Research Station, South Africa

- Engineered a smart camera system using Raspberry Pi and OpenCV, enhancing wildlife monitoring efficiency through real-time image processing and optimized data capture.
- Developed a resilient data transmission protocol (Hybrid DTN / HDTN) and solar-powered system, ensuring reliable and sustainable operation in remote conservation areas.

Flight Analysis Website https://track-flights-web.vercel.app/

- Deployed dynamic website and ETL pipeline to ingest flight data from Miami International Airport and automated data collection into a cloud environment.
- Built a live dashboard that queries the flight database and displays arrivals, departures, delays, and airport performance metrics.

Encryption/Decryption Website https://mendyraul.github.io/pages/encryption.html

- Designed and built a user-friendly website enabling secure communication through private key encryption.

EDUCATION

Florida Atlantic University **May 2023**
B.S. Computer Science, Minor in Cybersecurity 3.8 GPA

- **Publication**: J. Accurso, **R. Mendy**, A. Torres, Y. Tang — **A ChatGPT-like Solution for Power Transformer Condition Monitoring** (IEEE ICMLA) https://ieeexplore.ieee.org/document/10459800

Tufts University **May 2019**
B.S. Community Health

LEADERSHIP

Captain, Argentina National Lacrosse Team | Buenos Aires, Argentina **May 2018 – Present**

- Played in the FIL World Lacrosse Championship in Netanya, Israel.